

Ex.no:5b	DATA VISUALIZATION USING MATPLOTLIB – SALES DATA

AIM

To visualize and analyze the E-Commerce dataset using Line Chart, Bar Chart, and Pie Chart with Matplotlib to understand unit price trends, country-wise comparisons, and product category distribution.

ALGORITHM

1. Import Pandas and Matplotlib libraries.
2. Upload and load the E-Commerce dataset into a DataFrame.
3. Convert the order_date column into date format.
4. Group the data by order date and calculate the average unit price.
5. Plot the average unit price over time using a Line Chart.
6. Group the data by country and calculate the average unit price.
7. Plot the country-wise average unit price using a Bar Chart.
8. Count the products according to category.
9. Plot the category distribution using a Pie Chart.
10. Add suitable titles, axis labels, legends, and display the visualizations

CODE

```

from google.colab import files

import pandas as pd

import matplotlib.pyplot as plt

uploaded = files.upload()

file_name = list(uploaded.keys())[0]

df = pd.read_csv(file_name)

print(df.head())


df["order_date"] = pd.to_datetime(
    df["order_date"],
    errors="coerce"
)

daily_price = (
    df.groupby("order_date")["unit_price"]
plt.figure(figsize=(12, 5))

```

```
plt.plot(
    daily_price.index,
    daily_price.values,
    marker="o"
)
plt.title("Average Unit Price Over Time")
plt.xlabel("Order Date")
plt.ylabel("Average Unit Price")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
country_avg = (
    df.groupby("country")["unit_price"]
    .mean()
    .sort_values(ascending=False)
    .head(10)
)
plt.figure(figsize=(10, 5))

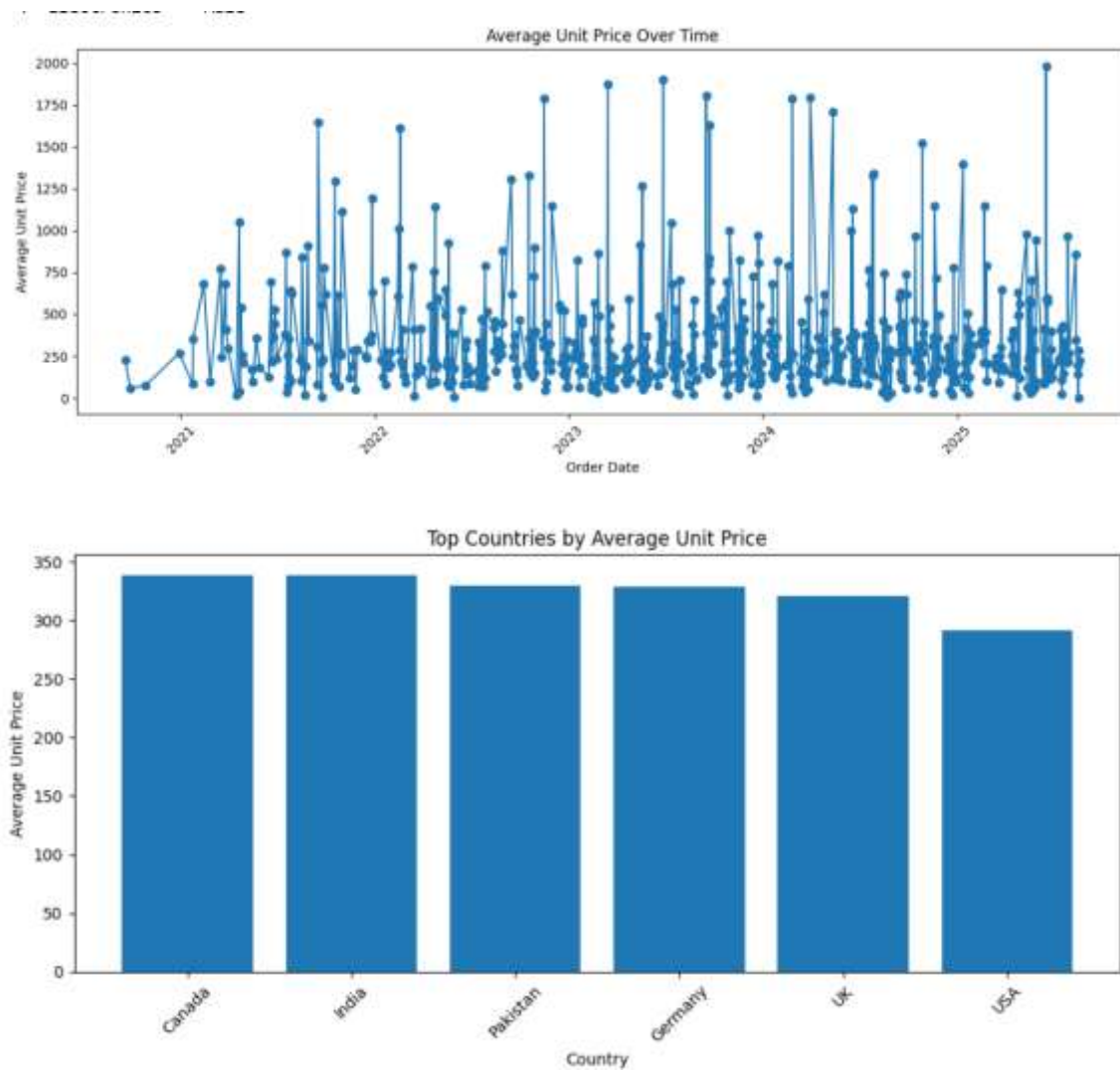
plt.bar(
    country_avg.index,
    country_avg.values
)
plt.title("Top Countries by Average Unit Price")
plt.xlabel("Country")
plt.ylabel("Average Unit Price")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
category_count = df["category"].value_counts()
plt.figure(figsize=(8, 8))
plt.pie(
    category_count.values,
    labels=category_count.index,
```

```

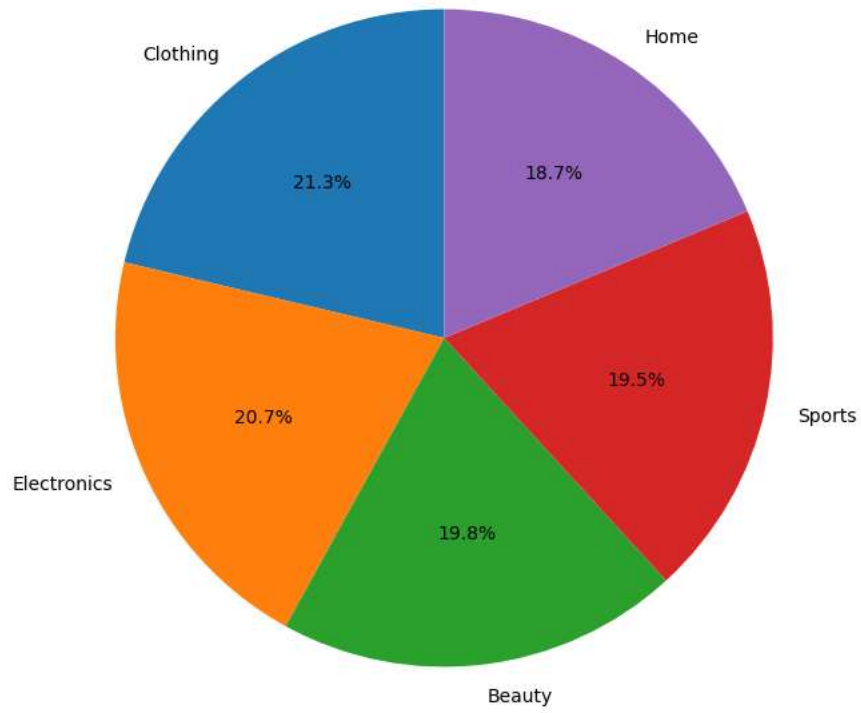
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Product Category Distribution")
plt.show()

```

OUTPUT



Product Category Distribution



RESULT